The Symmetry Tools of Particle Physics

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October 27, 2026

1 Abstract

Symmetry is a fundamental aspect of physics, connecting many of the most foundational equations and their solutions to each other using a mathematically rigorous backing. Its definition is simple: invariance under transformation. However, its implications are far-reaching. In this paper, we will use the basic axioms of special relativity to determine the representations of the Lorentz (and later the Poincaré) group to model transformations, symmetries, and objects in physics. These groups are the set of all transformations permitted by our symmetry constraints, that is, the isometry group of the Minkowski Space (preserves the Minkowski metric). From this, most equations in physics are derived, including the unification of the gauge symmetries. While not explicitly carving out these equations in this paper, we will set the framework in motion for interaction theory and all of theoretical particle physics.

2 Elementary Particles and Fundamental Forces

Let us quickly cover a shortened synopsis of our understanding of particle physics. There are two major categories for elementary particles: fermions and bosons. Fermions are the particles responsible for matter, and bosons are the particles responsible for the forces of nature (e.g. atoms are composed of fermions; however, bosons (specifically photons) are responsible for the electromagnetic force). There can never be two fermions in precisely the same state as provided by the Pauli Exclusion Principle, but there may exist infinitely many bosons.

Presently, we are only aware of four foundational forces being: the electromagnetic force (mediated by photons), the weak force (mediated by massive W^+, W^-, and Z^- bosons), the strong force (mediated by massless gluons), and gravity (which MIGHT be mediated gravitons). Note, every force is closely related to a symmetry. The process of spontaneous symmetry breaking is the cause of the masses of all elementary particles through the coupling of the Higgs-boson. Each corresponding gauge symmetry has a corresponding charge (electric, isospin, and color, respectively).

The fundamental fermions have two subcategories: the quarks, which compose protons and neutrons, and leptons, which compose electrons and neutrinos. The difference is that quarks carry can color charge, which allows them to engage in strong force interactions. With each category, there exist three "generations" of each particle that have two particles attributed to it, as shown below:

	Generation 1	Generation 2	Generation 3	Elec Charge	Isospin	Color Charge
Quarks	Up Down	Charm Strange	Top Bottom	⅔ e ⅓ e	½ -½	√ √
Leptons	Electron-Neutrino Electron	Muon-Neutrino Muon	Tauon-Neutrino Tauon	0 -e	½ -½	X X

Figure 1: Table of Particle Generations

Different particles can be identified with different labels (of which we will derive); that is, in addition to mass and charge, there exists a more fundamental label called spin, which can be seen as a form of pseudo-angular momentum. Bosons carry integer spin, whereas fermions carry half-integer spin. The fundamental fermions (leptons and quarks) have a spin of $\frac{1}{2}$. Contrarily, the majority of bosons

have spin 1, with the exception being the Higgs-boson with spin 0. The final notable point is that for every particle there exists a corresponding antiparticle, carrying the same labels with opposite sign. For example, the electron has an antiparticle, the positron. Some particles, such as a photon, are their own antiparticles.

3 Symmetry in Physics

Symmetry is a simple concept in definition: the symmetry of some system is a property of said system that is preserved under some set of transformations. We are usually introduced to symmetry in a geometric context (e.g., symmetries of a square or sphere). However, symmetry is a useful tool outside geometry alone, it offers a simplification to many strenuous problems in physics. Take, for example, a ball being thrown in the air. We can find the time it takes to hit the ground by simply doubling the time it takes to reach its peak height, as the motion of the ball is anti-symmetric (same in magnitude, but opposite in direction).

In a similar sense, we use symmetry to discuss the laws of physics. We know from special relativity that the laws of physics must be consistent in all inertial frames of reference. This means that our natural laws have an invariance under a set of transformations. We go into this result in the Appendix, but I will cover the methodology here. We start with the Minkowski space, which is commonly known as space-time. Here, there are three possible ways to transform an object (two linear and one non-linear) being rotations, boosts, and translations.

These form what is known as the Poincaré group (a subset of which is the Lorentz group: the set of all linear transformations that preserve the Minkowski metric). This group is known as a Lie group: a group that is represented by a differentiable manifold that provides a natural model for all continuous symmetries. It is given as the semi-direct product $\mathcal{P} = \mathbb{R}^{1,3} \rtimes \mathrm{O}(1,3)$. We, using representation theory, can describe the various representations of this group to model transformations for different mathematical objects. These representations lay the framework for our understanding and theory behind particle physics.

4 Special Relativity

To begin developing our theory of symmetry, we must discuss Einstein's results from special relativity. The famous Michelson-Morley Experiment discovered the speed of light is invariant in all reference frames. From this Einstein proved several invaluable corollaries, mainly: physics is consistent in all inertial reference frames and that the speed of light is the "speed limit" of our universe. (We can also assume that our universe is homogeneous and isotropic, meaning that no matter the orientation or the location in space-time, the laws of physics are identical.)

4.1 The Invariant of Special Relativity

We can use the postulates of special relativity to derive the Minkowski metric, which allows us to determine the distance between events in spacetime. We can refer to physical events as "points" in the Minkowski space, which is a four-dimensional real vector space with three spatial and one temporal dimensions. By the laws of special relativity, all transformations connecting inertial reference frames must not change the Minkowski space. The invariant of special relativity is a value that is constant in all inertial frames of reference. It is given by:

$$(\Delta s)^2 = (c\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \implies (ds)^2 = c^2(dt)^2 - (dx)^2 - (dy)^2 - (dz)^2$$

This quantity is maintained regardless of the observer. Thus, since this quantity must be preserved, observers in varying reference frames must not perceive time identically (for some primed and unprimed observer $\Delta t \neq \Delta t'$). This is referred to as time dilation. Because this relativity is observer-dependent, we can establish a reference called proper times such that: $(\Delta s)^2 = (c\Delta\tau)^2$ where τ is proper time.

From this invariant, we can derive the upper speed limit of all objects in our universe that being:

$$\Delta s^2_{min} \to 0 = (c\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \implies c^2 = \frac{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}{\Delta t^2} = |\vec{v}|^2 \implies |\vec{v_{max}}| = c$$

This is a quite popular result from physics, but it is crucial to understanding our mathematical framework. From this observation, we can derive the Principle of Locality. It states that everything in physics is only influenced by its immediate surroundings, which is restricted to the speed at which light travels.

4.2 The Minkowski Metric

Before continuing, I would like to introduce/review the conventional notation adopted from Ricci calculus, as it allows us to generalize a framework for defining, manipulating, and understanding tensor fields. As such, it becomes invaluable to describe the curvature of spacetime. It is as follows

- Einstein Summation Notation: if an index occurs twice, a sum is implicitly assumed: $a_i b_i \implies \sum_{i=1}^{3} a_i b_i$
- Greek Indices: indices like μ, ν, or σ are always summed implicitly from 0 to 3: $a_\mu b_\mu \implies \sum_{i=0}^{3} a_i b_i$
- The Minkowski Metric: $\eta^{00} = 1, \eta^{11} = -1, \eta^{22} = -1, \eta^{33} = -1$, and for all $x \neq y, \eta^{xy} = 0$. Thus, it can be expressed as a single four-by-four matrix:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

- Vectors: in addition, it's conventional to introduce that $dx_\mu = \begin{pmatrix} dx_0 \\ dx_1 \\ dx_2 \\ dx_3 \end{pmatrix}$

With this notation set, we can rewrite our invariant as

$$(ds)^2 = \eta^{\mu\nu} dx_\mu dx_\nu$$

A physical interpretation of ds is that it is the "distance" between two events in spacetime. If we consider the three-dimensional Euclidean space, this distance function is given by: $(ds)^2 = dx_i \delta_{ij} dx_j$ or, more simply, the Pythagorean Theorem. The mathematical tool that tells us the distance between two infinitesimal separate points is called a metric. In Euclidean space, the metric is just the identity matrix (δ). The geometry of the spacetime of special relativity is encoded in the Minkowski metric η. The tool to compute length, in this case, is precisely our invariant.

We can define the inner product for our Minkowski space as following $x \cdot y = x_\mu y_\nu \eta^{\mu\nu}$, and consequently, our length is defined to be $x \cdot x = (ds)^2$. Further, we can write $x^\mu = \eta^{\mu\nu} x_\nu$. Hence, we can simplify our definition of the product of two vectors as $x \cdot y = x_\mu y^\mu = x^\nu y_\nu$.

4.3 Lorentz Transformations

We will now determine the ways in which we can transform our given frame of reference without violating the principles of special relativity, known as a Lorentz transformation. That is $(ds)^2 = (ds')^2 = dx'_\mu du'_\nu \eta^{\mu\nu} = dx_\mu du_\nu \eta^{\mu\nu}$. These transforms leave the quadratic form/ scalar product of the space invariant. Denoting a generic transformation, we can use $dx_\mu \to dx'_\mu = \Lambda^\sigma_\mu dx_\sigma$, where Λ^σ_μ denotes the transformation operator.

With this, we can derive the defining condition of the set of all Lorentz transformations. As $(ds)^2 = (ds')^2 = dx \cdot dx = dx' \cdot dx' = dx_\mu dx_\nu \eta^{\mu\nu} = dx'_\mu dx'_\nu \eta^{\mu\nu} = \Lambda^\mu_\sigma dx_\sigma \Lambda^\gamma_\nu dx_\gamma \eta^{\sigma\nu} \implies \Lambda^\mu_\sigma \eta^{\sigma\gamma} \Lambda^\nu_\gamma = \eta^{\mu\nu} = \Lambda^T \eta \Lambda$. The Minkowski metric $\eta^{\mu\nu}$ remains unchanged under a Lorentz transformation Λ. Examples of transformations that preserve this property, as we will see later, are rotations and boosts. These properties are unique as they preserve the "length" of vectors in our Minkowski space.

All physical laws must be covariant under Lorentz transformations because such laws must be consistent in all frames of reference (given the postulates of special relativity). However, I'd like to point out that there is no preferred frame of reference for these laws, so we must formulate them in a covariant manner. Hence, we will be using symmetry to derive these laws.

5 Symmetry Tools

This section will be the main focus of the project, with the ultimate and final goal of deriving the fundamental representations of the double cover of the Poincaré group, the fundamental symmetry group of spacetime. It is a tool needed to describe all elementary particles as each representation classifies a category of particles.

5.1 Lie Groups

Lie groups are studied within a greater branch of mathematics known as group theory. For our interests, let us formally reiterate the definition a group:

1. A group is a set equipped with some binary operation–say the operation:*–that is closed under said operation (i.e. two elements when "combined" with said operation yields an element in said group: $\forall g_1, g_2 \in G, \exists g_3 \in G, g_1 * g_2 = g_3$)
2. In our set, we must contain some identity such that $\exists e \in G, g * e = e * g = g, \forall g \in G$
3. Every element in our set must have an inverse, an element that, when combined with the original element, will yield the identity element. That is $\forall g \in G, \exists g^{-1}, g^{-1} * g = g * g^{-1} = e$.
4. The group operation must be associative, meaning $\forall g_1, g_2, g_3 \in G, g_1 * g_2 * g_3 = (g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$

A Lie group, in particular, is a group that is also a differentiable manifold such that group multiplication and taking inverses are also differentiable. Moreover, the group operation $\circ$ induces a differentiable map from the manifold onto itself. This implies that there is precisely a single distinguishable Lie group for each Lie algebra (described in the next part) given by Lie's Third Theorem. It provides a natural model for continuous symmetries and serves as the foundational basis for our understanding of the Poincaré group, as we will see much later.

5.2 Lie Algebras

Lie Theory focuses on describing continuous symmetries as groups. Because of this, such groups contain infinitely many continuous symmetry transformations that can be continuously parameterized by one or several parameters (e.g., symmetries of a square are non-continuous as it only is rotations by multiples of 90 degrees, but a circle is symmetric for a rotation by any real-value angle).

Mathematically, with the identity transformation denoted I, an element arbitrarily close to the identity is denoted $g(\epsilon) = I + \epsilon X$, where X is the generator of our transformation and ϵ is some small value. Repeating this transformation often, we have $h(\theta) = (I + \epsilon X)^k$, where k denotes how often we repeat this transformation. If θ denotes some finite transformation parameter and N is some large number, we can rewrite $g(\epsilon)$ as $g(\theta) = I + \frac{\theta}{N}X$. Since we are considering infinitesimal transformations, we have it is, in actuality, $g(\theta) = \lim_{N\to\infty} I + \frac{\theta}{N}X$, and likewise, $h(\theta)$ is rewritten to be $h(\theta) = \lim_{N\to\infty}(I + \frac{\theta}{N}X)^N = e^{\theta X}$.

To determine the generator X of some given transformation, we can differentiate $h(\theta) = e^{\theta X}$ with respect to θ to get $\frac{dh(\theta)}{d\theta} = Xe^{\theta X}$. At, $\theta = 0$, we get $X = \frac{dh(\theta)}{d\theta}|_{\theta=0}$.

For some Lie group G, its Lie Algebra, $\mathfrak{g}$, is the tangent space of G at the identity element e. In other words, for matrix Lie groups (the ones we are concerned with) $G \subseteq GL(n)$, the Lie algebra $\mathfrak{g}$ consists of all nxn matrices X such that the matrix exponential $e^{tX} \in G, \forall t \in \mathbb{R}$. Our Lie algebra is equipped with a binary operation known as the Lie Bracket, which is an alternating bilinear map satisfying $[\,,\,] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$; $[A, B] = AB - BA$.

We can connect the combination rule of Lie Algebra to the combination rule of its Lie Group, using the Baker-Campbell-Hausdorff formula, given by $e^X \circ e^Y = e^{X+Y-1/2[X,Y]+1/12([X,[X,Y]]+[Y,[Y,X]])\cdots}$. We have two elements with generators, X and Y; the third is the combination of the two, which is represented by a long alternating sum of Lie brackets. The Lie bracket is

1. Alternating: $[x, x] = 0, \forall x \in \mathfrak{g}$
2. Anti-commutative: $[x, y] = -[y, x]$

3. Bilinear: $[ax, y] = a[x, y] = [x, ay]; [x+x', y] = [x, y]+[x', y]; [x, y+y'] = [x, y]+[x, y']; \forall x, x', y, y' \in \mathfrak{g}, \forall a \in \mathbb{R}$

4. Preserves the Jacobi Identity: $([x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0)$.

This definition reveals a more natural connection between Lie groups. As mentioned earlier, there is precisely a single distinguishable Lie group for each Lie algebra. This simply-connected group maps to all other groups with the same Lie algebra (but not vice versa). This group is known as the covering group. This becomes invaluable as we analyze rotational symmetries, the Lorentz group, and, most importantly, the Poincaré group.

5.3 Representation Theory

A key determinant of group theory is that it can describe any abstract object without the need for some concreteness. Therefore, it is beneficial to abstract any group when dealing with theoretical circumstances.

Another important idea in group theory is that a single group may act on many different possible "kinds" of objects. This significance inspired the emergence of a field in mathematics known as representation theory. A representation is a group homomorphism from a group to a linear transformation of some vector space V. That is $g \mapsto R(g)$, where $R(g)$ is a linear map $R(g) : V \to V$.

Representations identify with each element of a group, a linear transformation on V. Though a representation is definitionally a map, we will instead associate the title with a set of matrices for the sake of simplicity, as is common in the field. A result of this is that we can use representation theory to understand group action on differing vector spaces.

Before continuing, however, I would like to point out that we should introduce concepts that develop our understanding of the hierarchy of representations:

- Similarity Transformations: Given a representation $R(g)$ and an invertible matrix $S \in \mathrm{Aut}_{\mathbb{F}}(V)$. Applying a similarity transform on $R(g)$ by S, $R(g) \to R'(g) = S^{-1}R(g)S$, yields another representation (which is observed trivially). This means we can transform representations with non-singular matrices to obtain "nicer" matrices.

- Invariant Subspaces: For some representation R of a group G on a vector space V. We have that some subspace $V' \subseteq V$ is invariant if and only if for $v' \in V, R(g)(v') \in V', \forall g \in G$. We can define a subrepresentation R' of G on V', by $R'(g)v = R(g)v, v \in V'$.

- Irreducible Representations: An irreducible representation is a representation of a group G on a vector space V, which has no non-trivial invariant subspaces. These are "fundamental" representations as they are not composed of smaller representations; hence, we can build all representations from the set of all irreducible representations.

A question soon arises: Since there exist many possible representations for each Lie group, how do we know the "natural" representation? We answer this dilemma by introducing the Casimir element C, which is built from the generators of a given Lie algebra and commutes with every generator of a Lie group.

Schur's Lemma tells us that for some irreducible representation $R : \mathfrak{g} \to \mathrm{GL}_{\mathbb{F}}(V)$ any linear operator $C : V \to V$ that commutes with all operators R(J), must be a scalar multiple of the identity operator. Thus, Casimir elements must give us linear operators with constant values for each representation; these scalar values allow us to classify these operators.

One final note: for every Lie group, some number of the corresponding generators are diagonalizable. We can utilize these diagonal generators to label the basis of V, where the eigenvectors of the diagonalized generator act as our basis and the eigenvalues our labels. This means if there were to exist a single generator able to be simultaneously diagonalized, each basis is identifiable with solely an eigenvalue. This will lead to the charge labels for elementary particles.

5.4 Rotations

5.4.1 SO(3) and SU(2)

With the pretext set, let us consider a subset of transformations that leave the length of any vector unchanged. When we consider this, the first transformations that come to mind are rotations. Let us

consider the two-dimensional case. The generalized two-dimensional rotation matrices are of the form:

$$R_\theta = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

These transformations form a group under composition (or if using matrices, matrix multiplication). Further, we can formulate a formal definition of this group: $\vec{v} \to \vec{v'} \implies \vec{v} \cdot \vec{v} = \vec{v'} \cdot \vec{v'}$. Let us call this transformation $\mathcal{O}$, such that $v \to \vec{v'} = \mathcal{O}\vec{v}$. Since $\vec{v'} \cdot \vec{v'} = \vec{v}^T\vec{v} = (\mathcal{O}\vec{v})^T(\mathcal{O}\vec{v}) = \vec{v}^T\mathcal{O}^T\mathcal{O}\vec{v} = \vec{v} \cdot \vec{v} \implies \mathcal{O}^T\mathcal{O} = Id$. This shows the preservation of the identity by our transformation.

Additionally, we can add the constraint on the group to have $\det(\mathcal{O}) = 1$. These two conditions, taken together, yield a group known as $SO(2)$, where "S" signifies special and "O" signifies orthogonal (the 2 represents the dimensionality of our space).

While this is one way to describe a rotation, in the vector space $\mathbb{R}^2$, we can also consider rotations using the set of unit complex numbers. A rotation by some angle, θ, can be represented as a complex number of length one by Euler's Formula. This definition of unitary complex numbers forms the group U(1) under the group operation of multiplication. The connection between our two groups is quite evident; however, let us formalize it.

As mentioned, we can express our unit complex numbers using Euler's formula (as all unit complex numbers lie on the complex unit circle): $R_\theta = e^{i\theta} = \cos(\theta) + i\sin(\theta)$, and given our definition, $R_\theta^* R_\theta = 1$. Since a complex number has two degrees of freedom and our set has one constraint, we are effectively left with theta as our degree of freedom. We can represent complex numbers as 2x2 matrices over the field, $\mathbb{R}$. We can define:

$$1_{\mathbb{C}} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } i_{\mathbb{C}} \to \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \text{ so we have}$$

$$R_\theta = \cos(\theta) + i\sin(\theta) \to \cos(\theta)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \sin(\theta)\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

This is our understanding of all two-dimensional rotation matrices. We can trivially show this mapping is bijective, but it is left out for brevity. Thus, we have an isomorphism between SO(2) and U(1).

Let us try to do something similar with the three-dimensional case. The standard method to rotate vectors in three dimensions is to use 3x3 rotation matrices. The standard generators for the group of all rotations are:

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}, R_y = \begin{pmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{pmatrix} R_z = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

However, unlike our two-dimensional description of rotations, three-dimensional rotations cannot be expressed with individual complex numbers, so we need to find a comparable set of numbers to generalize rotations in three-space. To do so, we resort to using the four-dimensional group, the quaternions, implementing a constraint to reduce the degrees of freedom (four generators and one restriction).

The set of quaternions is analogous to the complex numbers with greater degrees of freedom. Rather than an element of the form $z = a + bi; a, b \in \mathbb{R}$, an element in this group would be of the form $q = a + bi + cj + dk; a, b, c, d \in \mathbb{R}$, with i,j,k satisfying $i^2 = j^2 = k^2 = -1$. Because of this property, our set is a non-abelian ring with the added structure that every element has an inverse. We can define the set of unit quaternions as the elements of the satisfying $q^\dagger q = 1 = (a - bi - cj - dk)(a + bi + cj + dk) = a^2 + b^2 + c^2 + d^2 = 1$. Precisely as the unit complex numbers form a group under multiplication, our unit quaternions do as well.

We can now represent each of the four units with a matrix like we have for the two-dimensional case. There are two ways to do so, with 2x2 complex matrices or 4x4 real matrices, but we will proceed with the former. It is as follows:

$$1_{\mathbb{H}} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, i_{\mathbb{H}} \to \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, j_{\mathbb{H}} \to \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, k_{\mathbb{H}} \to \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \text{ so for some quaternion,}$$

$$q = a + bi + cj + dk, \text{ we have } q \to q' = \begin{pmatrix} a + di & b + ci \\ -b + ci & a - di \end{pmatrix}$$

From this, $\det(q) = a^2 + b^2 + c^2 + d^2$; as from our conditions for the unit quaternions, this means $\det(q') = 1$. Hence, our mapped quaternions further satisfy the unitary conditions. So we have that our matrices form the group formally known as SU(2), or the Special Unitary group of dimension two.

With this, let us draw the connection between SO(3) and SU(2), though it is not as explicit as that for U(1) and SO(2). Let us define some pure quaternion $v \equiv ai + bj + ck$. We can use this as our vector in three-space. Using our matrix representation of quaternions, we have:

$$v \to \vec{v'} = x\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + y\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} + z\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} zi & x+yi \\ -x+yi & -zi \end{pmatrix}$$

Trivially, we have that $\det(\vec{v'}) = x^2 + y^2 + z^2$. If we want to consider the set of transformations that preserve the length of a vector, we must restrict our focus to matrices with unit determinants. However, simply multiplying a pure quaternion (vector) by some unit quaternion to rotate our vector won't always yield another pure quaternion. Thus, we modify the orientation of our vector by $v' = qvq^{-1}$, where q is our unit quaternion and v our vector. This will preserve the structure of our pure quaternion.

Let us observe a rotation of the vector $v = 1i+0j+0k$ about the z-axis by some angle θ. One would initially believe that the corresponding unit quaternion for our rotation is $R_\theta = \cos(\theta)+k\sin(\theta)$. This in its corresponding matrix form would be:

$$R_\theta = \begin{pmatrix} \cos(\theta)+i\sin(\theta) & 0 \\ 0 & \cos(\theta)-i\sin(\theta) \end{pmatrix} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \text{ so } R_\theta^{-1} = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

Applying our transformation:

$$\vec{v'} = R_\theta \vec{v} R_\theta^{-1} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix} = \begin{pmatrix} 0 & \cos(2\theta)+i\sin(2\theta) \\ -\cos(2\theta)+i\sin(2\theta) & 0 \end{pmatrix}$$
$$\implies v' = \cos(2\theta)i + \sin(2\theta)j + 0k$$

This means our process does indeed rotate our vector, but it is by a rotation double that of our desired angle. Therefore, we define $\phi \equiv 2\theta$ such that ϕ is the angle by which we rotate our vector. Our modified definition for the corresponding unit quaternion is $R_z = \cos(\frac{\phi}{2}) + \sin(\frac{\phi}{2})$. Applying this transformation, we have

$$\vec{v'} = R_\theta \vec{v} R_\theta^{-1} = \begin{pmatrix} e^{i\phi/2} & 0 \\ 0 & e^{-i\phi/2} \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} =$$
$$\begin{pmatrix} 0 & \cos(\phi)+i\sin(\phi) \\ -\cos(\phi)+i\sin(\phi) & 0 \end{pmatrix} \implies v' = \cos(\phi)i + \sin(\phi)j + 0k.$$

This means that the identifications we have made are not injective because for a vector rotation, there exist two unit quaternions that we can use to describe it. It is for this reason that we consider the group SU(2) to be a double cover of the group SO(3). While this seems like a technicality now, it is the foundation of the concept of spin in particle physics. To understand this concept, we consider the Lie Algebra of $SO(3)$ and $SU(2)$ to fully contextualize this idea of covering groups.

5.4.2 The Generators and Lie Algebra of SO(3)

As we know, the defining conditions of the group SO(3) are $O^T O = 1$ and $\det(O)$. Given the previous section, we can express any element in SO(3) as $O = e^{\theta J}$. From this, we know $1 = O^T O = (e^{\theta J})^T (e^{\theta J}) \implies J^T + J = 0$. Moreover, using the second equation, $\det(O) = 1 = e^{\theta \mathrm{tr}(O)} \implies \mathrm{tr}(J) = 0$. With this, we can choose three linearly independent matrices to act as our generators; the following matrices satisfy this definition:

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, J_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

These matrices form a basis for the generators of the group, so any generator of this group can be written as a linear combination of the set basis generators. We introduce the Levi-Civita symbol to express these generators compactly:

$(J_i)_{jk} = -\epsilon_{ijk}$, where j,k denote the components of the generator J_i, so for example,

$$(J_1)_{jk} = -\epsilon_{1jk} \rightarrow \begin{pmatrix} (J_1)_{11} & (J_1)_{12} & (J_1)_{13} \\ (J_1)_{21} & (J_1)_{22} & (J_1)_{23} \\ (J_1)_{31} & (J_1)_{32} & (J_1)_{33} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

Let us observe the rotation transformations that would result from the generators. For J_1, $R_1 = e^{\theta J_1}$. Further, we have that

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, J_1^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, J_1^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, J_1^4 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\implies J_1^{2n} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & (-1)^n & 0 \\ 0 & 0 & (-1)^n \end{pmatrix} = (-1)^{n-1} J_1^2 \text{ and } J_1^{2n-1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & (-1)^n \\ 0 & (-1)^{n+1} & 0 \end{pmatrix} = (-1)^{n-1} J_1$$

We can evaluate the exponential function:

$$R_1 = e^{\theta J_1} = \sum_{n=0}^{\infty} \frac{\theta^n J_1^n}{n!} = \sum_{n=0}^{\infty} \frac{\theta^{2n+1}}{(2n+1)!}(-1)^{n-1} J_1^2 + \sum_{n=0}^{\infty} \frac{\theta^{2n}}{(2n)!}(-1)^{n-1} J_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}$$

We can repeat this process for the other linearly independent generators. We see that the explicit matrix forms of our generators are the rotations given by real 3x3 matrices about our coordinate axes. From this, we can more directly compute the Lie bracket relations between our basis generators through brute force. We see that $[J_i, J_j] = \epsilon_{ijk} J_k$. Conventionally, we desire our generating matrices to be Hermitian such that $e^{i\theta J}$ is unitary, making the eigenvalues of our generators real. Therefore, our new generators are

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix}, J_2 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, J_3 = \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

5.4.3 The Generators and Lie Algebra of SU(2)

When discussing the rotations in three dimensions, we found that SU(2) is a double cover of SO(3). So, let us explore the Lie algebra of SU(2) to understand this connection. As a refresher, SU(2) is the group of 2x2 unitary matrices with unit determinants: $U^\dagger U = UU^\dagger = 1$ and $\det(U) = 1$. Writing these conditions with generators, we have $U^\dagger U = 1 = (e^{i\theta J_i})^\dagger (e^{i\theta J_i}) = (e^{-i\theta J_i^\dagger})(e^{i\theta J_i}) \implies J_i^\dagger = J_i$. Further, $\det(U) = \det(e^{-i\theta J_i}) = 1 \implies \mathrm{tr}(J_i) = 0$. Hence, our generators are Hermitian and traceless. A basis for these matrices is given by

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These are the Pauli matrices, and every 2x2 traceless Hermitian matrix can be written as a linear combination of these three matrices. Explicitly writing our generators with the Lie bracket yields $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$. To get rid of the 2 in the expression, we slightly modify our generators such that $J_i = \frac{1}{2}\sigma_i$. So, our new Lie algebra is $[J_i, J_j] = i\epsilon_{ijk} J_k$ and the generators are

$$J_1 = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, J_2 = \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}, J_3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

Note that this is the same Lie bracket relation as we defined for SO(3). Thus, there does exist an underlying condition between SO(3) and SU(2) in a similar way as the connection between SO(2) and U(1). Since SU(2) is diffeomorphic to the three sphere, it is the covering group of all other groups belonging to that Lie algebra. The map from $SU(2) \rightarrow SO(3)$ identifies two points in SU(2) to a single point in SO(3), so in a sense SO(3) is half of SU(2). It might seem non-conventional to think of a rotation in three dimensions as an element of SU(2) rather than SO(3), but its role as the covering group is much more fundamental in our theory.

5.5 SU(2)

For this section, we will be analyzing the various nth-dimensional representations of the SU(2) group to better understand its role as a subgroup of the Lorentz group $\mathcal{O}(1,3)$.

5.5.1 The Finite-Dimensional Irreducible Representations of SU(2)

As mentioned in the overview of Representation Theory, Casimir operators are built from a given set of generators that allow us to interpret the representation of a given Lie algebra. They, by construction, have the special property that allows them to commute with every generator of our Lie algebra. For $\mathfrak{su}(2)$, there exists one such operator which we identify as $J^2 = J_1^2 + J_2^2 + J_3^2$.

To highlight this property, let us consider the two-dimensional case:

$$J^2 = J_1^2 + J_2^2 + J_3^2 = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}^2 + \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}^2 + \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}^2 = \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{3}{4} \end{pmatrix}$$

This illustrates that for some set of generators given by some representation, we can calculate the corresponding Casimir operators mechanically. Acting the Casimir operator on some vector in our vector space, we obtain a number that we use as our label. We additionally define Cartan operators as the generators that form a basis for the Cartan subalgebra, a maximal commuting subalgebra of our Lie algebra, which can be simultaneously diagonalized. Because of this, it can be used to label the various elements of the vector space that our groups act on. In summary, Casimir operators provide labels to distinguish one representation from another, while Cartan elements provide distinguishing labels within a representation.

Let us look at the Lie algebra of $SU(2)$, $\mathfrak{su}(2)$. For this Lie algebra, there exists only one element that is diagonalized at a time, given by the fact that no two generators commute. By convention, we choose J_3 to be our diagonal generator; as in quantum mechanics, we use the z-component to define angular momentum states. It is given by:

$$J_3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

For each representation, the eigenvectors of the diagonal generator form the basis for the vector space our representation acts on. We, thus, label each basis vector by two numbers: b, given by the value of the quadratic Casimir operator J^2 of our representation, and m, the value obtained when acting our diagonal generator J_3 on our basis vectors. That is:

$$J^2 |b, m\rangle = b |b, m\rangle ; J_3 |b, m\rangle = m |b, m\rangle$$

In the case of the two-dimensional representation, our basis vectors in this notation are $|\frac{3}{4}, \frac{1}{2}\rangle, |\frac{3}{4}, -\frac{1}{2}\rangle$. This, however, is only one representation of $SU(2)$; to explore the various irreducible, finite-dimensional representations, we will introduce the ladder operators.

5.5.2 Ladder Operators

We define the ladder operators as $J_+ = J_1 + iJ_2$ and $J_- = J_1 - iJ_2$. These operators obey: $[J_3, J_+] = J_+, [J_3, J_-] = -J_-, [J_+, J_-] = 2J_3$. Let us now investigate the effect of these operators on our eigenvectors $|b, m\rangle$:

$$\begin{aligned} J_3(J_\pm |b, m\rangle) &= J_3(J_\pm |b, m\rangle) + J_\pm J_3 |b, m\rangle - J_\pm J_3 |b, m\rangle \\ &= J_\pm m |b, m\rangle + [J_3, J_\pm] |b, m\rangle = (m \pm 1) J_\pm |b, m\rangle \end{aligned}$$

It is evident that $J_\pm |b, m\rangle$ is an eigenvector of J_3, but with an eigenvalue of $m \pm 1$, so $J_\pm |b, m\rangle \equiv C |b, m \pm 1\rangle$ and $J_3 |b, m \pm 1\rangle = (m \pm 1) |b, m \pm 1\rangle$. Thus, the reasoning behind the title of ladder operators becomes more apparent: these operators allow us to traverse "up" and "down" irreducible representations. However, since our vector space is finite-dimensional and our eigenvectors with differing eigenvalues are always linearly independent, there must be an eigenvector with a maximal and minimal eigenvalue (which we label j and k, respectively). That is $J_+ |b, j\rangle = 0, J_- |b, k\rangle = 0$.

We can explicitly calculate the relationship between our maxima eigenvalues and the label b by using $J_- J_+ = J^2 - J_3^2 - J_3$. Applying $J_- J_+$ and $J_+ J_-$ to our maxima, we get $J_+ J_- |b, j\rangle = (b - j^2 - j) |b, j\rangle =$

$0 \implies b - j^2 - j = 0, b = j(j+1)$ and $J_+ J_- |b,k\rangle = (b - k^2 + k)|b,k\rangle = 0 \implies b - k^2 + k = 0, b = -k(-k+1)$. This shows that, in actuality, $k = -j$. If we consider the distance between the maximum and minimum eigenvalue of J_3, we get that our j value must be some integer divided by two, since our eigenvalues are integer-spaced.

With this, we can explicitly find all the constructions of possible finite-dimensional representations of $\mathfrak{su}(2)$:

j	m	b
0	0	0
1/2	1/2, -1/2	3/4
1	-1, 0, 1	2
...	...	...

We have that each half-integer representation corresponds to a dimensional representation of our group: j=0 is the one-dimensional representation, j=1/2 is the two-dimensional representation, etc. Note that the values corresponding to j=1/2 were the ones we had explicitly calculated at the beginning of this section. We can now derive our Casimir operator, C.

To make our work easier, it is best if we always choose a normalized basis, such that $|b,m\rangle^\dagger |b,m\rangle = 1$. If we act $J_\pm$ onto our normalized basis, the result is not guaranteed to be normalized. This fact allows us to derive C. Acting J_+ onto our normalized basis, we get that:

$$(J_+|b,m\rangle)^\dagger (J_+|b,m\rangle) = |C|^2 |b,m\rangle^\dagger |b,m\rangle = |C|^2 = |b,m\rangle^\dagger J_+^\dagger J_+ |b,m\rangle =$$

$$(j(j+1) - m^2 - m) \implies C = \sqrt{(j(j+1) - m^2 - m)}$$

Conversely, for J_-, the $\tilde{C}$ is $\sqrt{(j(j+1) - m^2 + m}$. These results make sense as if we were to apply J_+ to j or J_- to $k = -j$, we see that C and $\tilde{C}$ are equal to zero, respectively. With this, we are equipped to calculate explicit matrices for the various SU(2) representations.

5.5.3 The Representation of SU(2) in One Dimension

The smallest value for j is zero; this representation acts on a one-dimensional vector space. Thus, this representation must be trivial since the only one-dimensional generating matrix J_k that satisfies $[J_i, J_j] = i\epsilon_{ijk} J_k$ is the zero matrix. If we were to consider this as an explicit matrix in SU(2), it would be the identity matrix.

5.5.4 The Representation of SU(2) in Two Dimensions

The next value of j is one-half. This representation acts on a two-dimensional vector space and, therefore, must be two-dimensional. The generator J_3 has eigenvalues $\frac{1}{2}$ and $-\frac{1}{2}$, and the corresponding eigenvalues are $|\frac{3}{4}, \frac{1}{2}\rangle = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $|\frac{3}{4}, -\frac{1}{2}\rangle = \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$. With this, we can find the explicit matrix forms of the other SU(2) generators J_1 and J_2 by rewriting them using ladder operators:

$$J_1 = \frac{1}{2}(J_- + J_+) \implies J_1 |\frac{3}{4}, \frac{1}{2}\rangle = \frac{1}{2}|\frac{3}{4}, -\frac{1}{2}\rangle ; J_1 |\frac{3}{4}, -\frac{1}{2}\rangle = \frac{1}{2}|\frac{3}{4}, \frac{1}{2}\rangle$$

$$\implies J_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Repeating this process for J_2, we get that $J_2 = \frac{1}{2}\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$. These again are the same generators we derived previously, where $J_i = \frac{\sigma_i}{2}$, when deriving $\mathfrak{su}(2)$ in two-dimensions.

5.5.5 The Representation of SU(2) in Three Dimensions

Following the same methodology as we had done for the two-dimensional case, we yield the generator matrices as follows:

$$J_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, J_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, J_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

As we have shown, this methodology can derive the generating matrices for any representation. For our purposes, we will not derive any higher-order representations.

5.6 The Lorentz Group: $\mathcal{O}(1,3)$

In this section, we will be describing and deriving a well-known representation of the Lorentz Group, allowing us to derive its Lie algebra and other representations of the group. Arguably, the simplest representation is a four-dimensional vector space, so our representations act on our vector space as 4x4 matrices. Before proceeding, let us redefine the Lorentz group: it is the set of all transformations that preserve the inner product of the Minkowski Space (which we derived earlier) $x^\mu x_\mu = x^\mu \eta_{\mu\nu} x^\nu = x_0^2 - x_1^2 - x_2^2 - x_3^2$.

The reason why we call it $\mathcal{O}(1,3)$ is that the Minkowski Space's metric signature is (1,3). We notate Lorentz transformations as Λ. As we definied Λ before, $x^\mu \eta_{\mu\nu} x^\nu \to x'^\sigma \eta_{\sigma\rho} x'^\rho = (x^\mu \Lambda^\sigma_\mu)\eta_{\sigma\rho}(\Lambda^\rho_\nu x^\nu) = x^\mu \eta_{\mu\nu} x^\nu \implies \Lambda^\sigma_\mu \eta_{\sigma\rho} \Lambda^\rho_\nu = \eta_{\mu\nu} = \Lambda^T \eta \Lambda$. With this understood, let us analyze its results. Consider $det(\Lambda^T)det(\eta)det(\Lambda) = det(\eta) \implies det(\Lambda)^2 = 1$, so $det(\Lambda) = \pm 1$. Further, if $\mu = \nu = 0 \implies \Lambda^\sigma_0 \eta_{\sigma\rho} \Lambda^\rho_0 = \eta_{00} = -1 \implies (\Lambda^0_0)^2 - \sum(\Lambda^i_0)^2 = 1$, so $\Lambda^0_0 = \pm\sqrt{1 + \sum(\Lambda^i_0)^2}$

We therefore have four emergent categories of our Lorentz transformations: $L^\uparrow_+ \ (det(\Lambda) = 1, \Lambda^0_0 \geq 1), L^\downarrow_+ \ (det(\Lambda) = 1, \Lambda^0_0 \leq -1), L^\uparrow_- \ (det(\Lambda) = -1, \Lambda^0_0 \geq 1), L^\downarrow_- \ (det(\Lambda) = -1, \Lambda^0_0 \leq -1)$. This classification is significant as only $L^\uparrow_+$ can be generated through infinitesimal transformation (as $I \in \mathcal{O}(1,3)$ is solely in that category). The transformations within the other categories are obtained by applying the time-inversion and space-inversion operators to elements in $L^\uparrow_+$: given by

$$\Lambda_P = -\Lambda_T = \eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}$$

To contextualize this, $L^\uparrow_+ = (Id)(L^\uparrow_+); L^\downarrow_+ = \Lambda_T L^\uparrow_+; L^\uparrow_- = \Lambda_P L^\uparrow_+; L^\downarrow_- = \Lambda_P \Lambda_T L^\uparrow_+$. This shows a very important fact about these categories: they are not smoothly connected to one another. Our time-inversion and space-inversion transformations are discrete, so we have the subcategory that $L^\uparrow_+$ generates our other subcategories.

5.6.1 A Representation of the Lorentz Group

The representation of the Lorentz group we are concerned with is given by 4x4 real matrices. From this, we have sixteen parameters, ten constraints (given by the fact that the Minkowski metric is symmetric), and six free parameters. Thus, we must search for six generating matrices for our group. A good launching platform is given by the fact that we have been studying groups generated by three such matrices: the rotations in three-space. They satisfy our definition of Lorentz transformations, and all three fall into the subcategory $L^\uparrow_+$.

The corresponding Lorentz transformations are given by $\Lambda_{rot} = \begin{pmatrix} 1 & \\ & R_{3x3} \end{pmatrix}$ and the corresponding generators are similar to those for three-space $J_i = \begin{pmatrix} 0 & \\ & J_i^{3-dim} \end{pmatrix}$. To investigate transformations involving temporal and spatial components, we will start with infinitesimal transformations: $\Lambda^\mu_\rho \approx \delta^\mu_\rho + \epsilon K^\mu_\rho$ where δ^μ_ρ is the Kronecker delta. For $\Lambda^\mu_\rho \eta_{\mu\nu} \Lambda^\nu_\sigma = \eta_{\rho\sigma} \approx (\delta^\mu_\rho + \epsilon K^\mu_\rho)\eta_{\mu\nu}(\delta^\nu_\sigma + \epsilon K^\nu_\sigma) = \eta_{\rho\sigma} \implies K^\mu_\rho \eta_{\mu\sigma} + K^\nu_\sigma \eta_{\rho\nu} = 0$, so $K^T \eta = -\eta K$.

This condition for the generators of the transformation, involving spatial and temporal components, reveals a new class of generators (other than rotations) known as a boost. A boost is a change into a coordinate system that moves with a different, constant velocity compared to the coordinate system. We can create a basis of linearly independent generators, which we can use to generate any boost:

$$K_x = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}; K_y = \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}; K_z = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}$$

From our understanding of Lie Theory and matrix exponentials, we can calculate the explicit transformation matrices, using $e^{i\theta K_i}$:

$$\Lambda_x = \begin{pmatrix} \cosh(\phi) & -\sinh(\phi) & 0 & 0 \\ -\sinh(\phi) & \cosh(\phi) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}; \Lambda_y = \begin{pmatrix} \cosh(\phi) & 0 & -\sinh(\phi) & 0 \\ 0 & 1 & 0 & 0 \\ -\sinh(\phi) & 0 & \cosh(\phi) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix};$$

$$\Lambda_z = \begin{pmatrix} \cosh(\phi) & 0 & 0 & -\sinh(\phi) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh(\phi) & 0 & 0 & \cosh(\phi)) \end{pmatrix}$$

An arbitrary boost can be composed by combining these three matrices, given the group operation of multiplication. To understand the generators for the transformations of the other subcategories of the Lorentz group, we must apply Λ_P or Λ_T. By brute force, we find that $J_i \mapsto J_i$ and $K_i \mapsto -K_i$ for Λ_T and Λ_P.

5.6.2 The Lie Algebra of $L_+^\uparrow$

Now, using the explicit matrix form of the generators in $L_+^\uparrow$, we can derive the corresponding Lie algebra by brute force computation: $[J_i, J_j] = i\epsilon_{ijk}J_k; [J_i, K_j] = i\epsilon_{ijk}K_k; [K_i, K_j] = -i\epsilon_{ijk}J_k$, where J_i are the generators for rotations and K_i is the generators of boosts. Thus, we have that a general Lorentz transformation is given by $\Lambda = e^{i(\vec{J}\cdot\vec{\theta}+\vec{K}\cdot\vec{\phi})}$.

Since $[J_i, K_j] \neq 0$, J_i and K_j don't commute with one another. While the rotation generators are closed under commutation, the boost generators are not. Hence, we can define two new operators that are closed under commutation: $N_i^\pm = \frac{1}{2}(J_i \pm iK_i) \implies [N_i^\pm, N_j^\pm] = i\epsilon_{ijk}N_k^\pm, [N_i^+, N_j^-] = 0$. These are the commutative relations that we have seen for the Lie algebra of $SU(2)$, and we have discovered that the "complexified" Lie algebra of $L_+^\uparrow$ consists of two copies of $\mathfrak{su}(2)$. We now have a derivation for the irreducible representation of the Lie algebra of $SU(2)$. Since the Lorentz group is not simply connected, we will find the irreducible representation of the covering group of the Lorentz group.

Each irreducible representation of $\mathfrak{su}(2)$ is labeled by the scalar value j of the respective Casimir operator. Hence, we can label the irreducible representations of the covering group of the Lorentz group with two integers/half-integers: j_1, j_2. We will look at the (j_i, j_j) representations for $j_i, j_j = 0, 1/2, 1, 3/2...$.

Before continuing, we will introduce one final operator $M_{\mu\nu} : J_i = \frac{1}{2}\epsilon_{ijk}M_{jk}, K_i = M_{0i} \implies [M_{\mu\nu}, M_{\rho\sigma}] = i(\eta_{\mu\rho}M_{\nu\sigma} - \eta_{\mu\sigma}M_{\nu\rho} - \eta_{\nu\rho}M_{\mu\sigma} + \eta_{\nu\sigma}M_{\mu\rho})$. We will use this to lead as a tool in our investigation of different representations of the complexified Lie algebra of the Lorentz group.

5.6.3 The (0,0) Representation

The lowest dimension representation is a trivial representation, because our vector space is one-dimensional for both copies of $\mathfrak{su}(2)$. Our generators are 1x1 matrices, and the only matrix fulfilling our requirements is the zero matrix. Thus, $N_i^+ = N_i^- = 0 \implies e^{iN_i^\pm} = 1$. Consequently, this is known as the scalar representation of the Lorentz group.

5.6.4 The $(\frac{1}{2}, 0)$ Representation

In this representation, we have access to the two-dimensional copy of $\mathfrak{su}(2)$ ($N_i^+ = \frac{\sigma_i}{2}$) and the one-dimensional representation for the other ($N_i^- = 0$). Using our defined N_i^+, N_i^-, we have that $J_i = iK_i, N_i^+ = iK_i = \frac{\sigma_i}{2} \implies K_i = \frac{i}{2}\sigma_i, J_i = \frac{1}{2}\sigma_i$. Consequently, a Lorentz rotation is expressed as $R_\theta = e^{i\vec{\theta}\cdot\vec{J_i}}$, and a boost is expressed as $B_\phi = e^{i\vec{\phi}\cdot\vec{K}}$.

From this, we can explicitly calculate matrices, but it has been omitted for brevity. One important result is that we have here 2x2 complex matrices, representing the Lorentz transformations. These matrices do not act on the four vectors of the Minkowski space as we have been led to believe. Rather, they act on two-component objects known as left-chiral spinors:

$$X_L = \begin{pmatrix} (X_L)_1 \\ (X_L)_2 \end{pmatrix}$$

We define specifically left-chiral spinors as the object upon which Lorentz transformations act on by the $(\frac{1}{2},0)$ representation. Spinors, as a general concept, are elements in a complex vector space that transform linearly when subjected to an infinitesimal rotation. However, unlike standard vectors, a spinor transforms to its negative when rotated by 360°. To rotate to its identity, one would need to rotate a spinor by 720°. For an explanation, our objects are spinors by virtue of the $\frac{1}{2}$ factor in $J_i = \frac{1}{2}\sigma_i$ and $K_i = \frac{i}{2}\sigma_i$.

Thus far, we have discussed the trivial representation, known as the scalar representation. The next rung on the hierarchy is known as the spinor representation, and it is given by the $(\frac{1}{2},0), (0,\frac{1}{2})$, $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ representations. We will discuss $(0,\frac{1}{2})$ in our next section.

5.6.5 The $(0,\frac{1}{2})$ Representation

This representation has a similar construction to that of the $(\frac{1}{2},0)$ representation. The key difference is that our one-dimensional representation is attributed to $N_i^+ = 0$, and the two-dimensional representation for $N_i^- = \frac{\sigma_i}{2}$. Therefore, we have that $J_i = -iK_i$, and since $N_i^- = \frac{1}{2}(J_i - iK_i) = -iK_i \implies K_i = \frac{i}{2}\sigma_i$. For this representation, we have that our rotation transformations are given by $R_\theta = e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}}$ and boost transformations are given by $B_\phi = e^{-\vec{\phi}\cdot\frac{\vec{\sigma}}{2}}$. Comparing this result with the $(\frac{1}{2},0)$ representation, we see that the rotation transformations are of identical form, but the boost transformations are inverses of each other (the exponent differs by a factor of -1). As we have for left-chiral spinors, we define the object that the $(0,\frac{1}{2})$ act on as right-chiral spinors:

$$X_R = \begin{pmatrix}(X_R)^1\\(X_R)^2\end{pmatrix}$$

In general, we call left-chiral (left-handed) spinors and right-chiral (right-handed) spinors Weyl spinors. We will go into this notation in the next section.

5.6.6 Van der Waerden Notation

We have explicitly seen the similarities between the $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ representations, so we will focus on a new type of notation known as Van der Waerden notation, which allows us to manipulate Weyl spinors. Left-chiral spinors are denoted as X_α, with a lower, undotted index. Conversely, right-chiral spinors are denoted as $X^{\dot\alpha}$, with an upper, dotted index.

There is an object known as the spinor metric, which allows for the conversion of a right-chiral spinor into a left-chiral spinor and a left-chiral spinor into a left-chiral spinor. We define it as $\epsilon^{ab} = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$. We can describe a transformation known as charge conjugation (which we will later analyze) as $X_L^C = \epsilon X_L^*$, where $*$ denotes complex conjugation. With this, we will analyze how X_L^C acts. For some boost transformation, we have:

$$X_L^C \to X_L^{'C} = \epsilon X_L^{'*} = (\epsilon)e^{\frac{\phi}{2}\sigma^*}(-\epsilon)(\epsilon)X_L^* = e^{-\frac{\phi}{2}\sigma}X_L^C$$

This means that X_L^C acts as a right-chiral spinor. Conversely, we can transform a right-chiral spinor into a left-chiral spinor by $X_L = \epsilon X_R^*$. This means that complex conjugation is the operation that allows us to transform a dotted index to an undotted index and vice versa. Further, it is important to understand that since ϵ is our metric, it is preserved through transformations.

Let us now analyze how a $X_{\dot\alpha}$ and X^α because these objects are needed to construct terms from spinors that don't change under Lorentz transformations. This will make more sense in a moment, but for now, let us observe how a generic Lorentz transformation affects our spinors

$$X_\alpha \to X'_\alpha = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}+\vec{\phi}\cdot\frac{\vec{\sigma}}{2}})_a^b X_b$$

Thus, a spinor with a lower, dotted index transforms as:

$$X^*_\alpha = X_{\dot\alpha} \to X'_{\dot\alpha} = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma^*}}{2}+\vec{\phi}\cdot\frac{\vec{\sigma^*}}{2}})_{\dot a}^{\dot b} X_{\dot b}$$

Similarly, for right-chiral spinor transformations:

$$X^{\dot\alpha} \to X'^{\dot\alpha} = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2}-\vec{\phi}\cdot\frac{\vec{\sigma}}{2}})_{\dot a}^{\dot b} X^{\dot b}$$

So, a spinor with an upper, undotted index transforms as:

$$(X^{\dot{\alpha}})^* = X^{\alpha} \to X'^{\alpha} = (e^{-i\vec{\theta}\cdot\frac{\vec{\sigma_*}}{2} - \vec{\phi}\cdot\frac{\vec{\sigma_*}}{2}})^{a}_{b} X^{b}$$

To be able to write products of spinors that do not change under Lorentz transformations, we need to understand one more property. Like with the scalar product of two vectors, we can express the spinor product with a transposed spinor and a standard spinor. This is hinted at by the conjugated Pauli matrices in the exponents of our transformations.

Given that the Hermitian conjugation of our Pauli matrices is $\sigma_i^\dagger = (\sigma_i^*)^T = \sigma_i$, we can observe that the transformation behavior of a transpose spinor with a lower, undotted index is opposite of a spinor with an upper, undotted index. Thus, $(X^{\alpha})^T X_{\alpha}$ is invariant under Lorentz transformations. Likewise, the same is true for $(X^{\dot{\alpha}})^T X_{\dot{\alpha}}$. Note, the same is not true for $(X^{\dot{\alpha}})^T X_{\alpha} = (X^{\alpha})^T X_{\alpha}$. Therefore, to get a Lorentz invariant term with right-chiral and left-chiral spinors, we must express them as $X_R^\dagger X_L$ or vice versa.

This provides further justification for the spinor metric: the invariant metric can be written as $(X_a)^T X^{\alpha} = X_a^T \epsilon^{ab} X_b$, which is of the same form of our Minkowski metric $x_\mu y^\nu = x_\mu \eta^{\mu\nu} y_\nu$. For housekeeping, we can express $\epsilon_{ab} = \epsilon^{ab}$, which is the metric that allows us to convert between lower, dotted spinors and upper, undotted spinors. Additionally, we can now write our transformation operators as one object Λ, where $X_R \to X'_R = X'^{\dot{a}} = \Lambda^{\dot{a}}{}_{\dot{b}} X^{\dot{b}}$ and $X_L \to X'_L = X'_a = \Lambda_a{}^b X_b$. Therefore, $\Lambda_{(\frac{1}{2},0)} = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2} + \vec{\phi}\cdot\frac{\vec{\sigma}}{2}}) := \Lambda_a{}^b$ and similarly $\Lambda_{(0,\frac{1}{2})} = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2} - \vec{\phi}\cdot\frac{\vec{\sigma}}{2}}) := \Lambda^{\dot{a}}{}_{\dot{b}}$.

5.6.7 The $(\frac{1}{2}, \frac{1}{2})$ Representation

For this representation, we have the two-dimensional representation for either copy of our $SU(2)$ group, N_i^+, N_i^-. While this may not seem trivial, since our N_i^+, N_i^- don't commute, our objects transform independently from one another. This object will thus have two indices given by $v_a^{\dot{b}}$, where, as we mentioned before, each index transforms independently. Thus, our object has four components since each representation is two-dimensional and both indices can take on two values $(\frac{1}{2}, -\frac{1}{2})$. Therefore, the object can be expressed as 2x2 matrices, but we can also enforce a four-component vector form.

A general 2x2 matrix has eight total parameters, four real and four imaginary. An important result from linear algebra is that we can express any complex matrix as the sum of a Hermitian and an anti-Hermitian matrix (each having four free parameters). Since we must transform Hermitian matrices to Hermitian matrices and vice versa, these matrices form an invariant subset. Further, we know that any two-dimensional arbitrary Hermitian matrix can be uniquely written as a linear combination of the Pauli matrices and the identity matrix.

Instead of examining $v_a^{\dot{b}}$ explicitly, we will be working with $v_{a\dot{b}}$, so we can interface with the Pauli matrices. We learned in the previous section that we can convert between the two objects by using ϵ. With this, and defining $\sigma^0 = I_{2\times 2}$, we can write $v_{a\dot{b}} = a_\nu \sigma^\nu_{a\dot{b}}$. Therefore, an arbitrary two-dimensional Hermitian matrix is given by:

$$v_{a\dot{b}} = \begin{pmatrix} a_0 + a_3 & a_1 - ia_2 \\ a_1 + ia_2 & a_0 - a_3 \end{pmatrix}$$

Now, let us consider a transformation of such an object. Since this representation is two copies of the two-dimensional representation of $SU(2)$, we transform our indices according to their corresponding representation, so:

$$v \to v' = v'_{a\dot{b}} = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2} + \vec{\phi}\cdot\frac{\vec{\sigma}}{2}})_a{}^c v_{c\dot{d}} ((e^{-i\vec{\theta}\cdot\frac{\vec{\sigma_*}}{2} + \vec{\phi}\cdot\frac{\vec{\sigma_*}}{2}})^{\dot{d}}{}_{\dot{b}})^T = (e^{i\vec{\theta}\cdot\frac{\vec{\sigma}}{2} + \vec{\phi}\cdot\frac{\vec{\sigma}}{2}})_a{}^c v_{c\dot{d}} (e^{-i\vec{\theta}\cdot\frac{\vec{\sigma}}{2} + \vec{\phi}\cdot\frac{\vec{\sigma}}{2}})^{\dot{d}}{}_{\dot{b}}$$

This transformation is easily verifiable to be Hermitian, but we will leave that proof out for the sake of brevity. While what we have now seems to be unimpressive, suppose we were transforming our object with a boost along the z-axis of our coordinate system:

$$v_{a\dot{b}} \to v'_{a\dot{b}} = (e^{\vec{\phi}\cdot\frac{\vec{\sigma_3}}{2}})^c_a v_{c\dot{d}} (e^{\vec{\phi}\cdot\frac{\vec{\sigma_3}}{2}})^{\dot{d}}_{\dot{b}} = \begin{pmatrix} e^{\phi}(a_0 + a_3) & a_1 - ia_2 \\ a_1 + ia_2 & e^{-\phi}(a_0 - a_3) \end{pmatrix} = \begin{pmatrix} a'_0 + a'_3 & a'_1 - ia'_2 \\ a'_1 + ia'_2 & a'_0 - a'_3 \end{pmatrix}$$

$$\implies a'_0 + a'_3 = e^{\phi}(a_0 + a_3) = (\cosh(\phi) + \sinh(\phi))(a_0 + a_3),$$
$$a'_0 - a'_3 = e^{\phi}(v_0 - v_3) = (\cosh(\phi) - \sinh(\phi))(a_0 - a_3)$$

$$\implies \nu_0' = \cosh(\phi)\nu_0 + \sinh(\phi)\nu_3; \nu_3' = \sinh(\phi)\nu_0 + \cosh(\phi)\nu_3$$

Thus, we have:

$$\begin{pmatrix} \nu_0' \\ \nu_1' \\ \nu_2' \\ \nu_3' \end{pmatrix} = \begin{pmatrix} \cosh(\phi) & 0 & 0 & \sinh(\phi) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh(\phi) & 0 & 0 & \cosh(\phi) \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \\ \nu_4 \end{pmatrix}$$

This holds for all Lorentz transformations. What this means is that **the** $(\frac{1}{2}, \frac{1}{2})$ **representation IS the vector representation**. This is the same 4x4 matrix representation that we have derived earlier. Thus, we can express our transformation laws through vectors rather than spinors or other mathematical objects (and it's preferable to 2x2 matrices as matrix-vector multiplication tends to be simpler). While this is a pragmatic discovery, we cannot use four-vectors to describe every physical system, since spinors are the more "fundamental" object. Nevertheless, now we have a better understanding of the relationship between spinors and vectors, adding credibility to spinors being considered "the square root of vectors."

With this, let us look back and resume our study of the parity and time-reversal operators, allowing us to access the entirety of the Lorentz group, not just $L_+^\uparrow$. Note, technically, claiming the universe is symmetric under these transformations is a gross oversimplification. This statement was disproved in the paper "Question of Parity Conservation in Weak Interactions"- Yang and Lee. But, we will be proceeding with the cautious perspective that it is maintained.

5.6.8 Spinors and Parity

Recall from earlier that under parity transformations $J_i \xrightarrow{P} J_i$ and $K_i \xrightarrow{P} -K_i$. Looking at the generators $N^\pm = \frac{1}{2}(J_i \pm iK_i)$, $N_i^+ \to N_i^-$ and $N_i^- \to N_i^+$ under this transformation. Consequently, under parity transforms, $(\frac{1}{2}, 0) \longleftrightarrow (0, \frac{1}{2})$. This mimics how under parity transformations left-handed and right-handed coordinate systems swap; it is from this that our chiral spinors derive their namesakes. Let us look deeper at this. Rotations are preserved by transformations, but boost transformations are inverted; that is:

$$(\Lambda_{\vec{K}})_{(\frac{1}{2},0)} = e^{\vec{\phi}\cdot\vec{K}} \longleftrightarrow e^{-\vec{\phi}\cdot\vec{K}} = (\Lambda_{\vec{K}})_{(0,\frac{1}{2})}$$

If we want to describe a physical system that is invariant under parity transformations, we must consider an object comprised of right-chiral and left-chiral spinors. This singular object is commonly known as the Dirac spinor:

$$\psi = \begin{pmatrix} \chi_L \\ \varphi_R \end{pmatrix}$$

A Dirac spinor ψ consists of two Weyl spinors; we clarify Weyl spinors to generalize these spinors to avoid bias towards discussing left-chiral and right-chiral spinors (they aren't necessarily fundamental). The type of Dirac spinor expressed with a left-chiral and right-chiral spinor is known as a Majorana spinor. Note, a Dirac spinor, while composed of four components, is not a four-vector as it transforms differently. In fact, it transforms with respect to the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation of the Lorentz group.

This is one type of reducible representation known as an off-diagonal direct sum of the Lorentz group. It is named as such since it takes the form $(x, y) \oplus (y, x), x \neq y$. It operates over $\mathbb{C}$, so the representation permits the use of linear operators over $\mathbb{R}$. We can write such a transformation as:

$$\psi \to \psi' = \Lambda_{(\frac{1}{2},0)\oplus(0,\frac{1}{2})}\psi = \begin{pmatrix} \Lambda_{(\frac{1}{2},0)} & 0 \\ 0 & \Lambda_{(0,\frac{1}{2})} \end{pmatrix} \begin{pmatrix} \chi_L \\ \varphi_R \end{pmatrix}$$

Note, $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ is a reducible representation. Conversely, four vectors transform according to the $(\frac{1}{2}, \frac{1}{2}) = (\frac{1}{2}, 0) \otimes (0, \frac{1}{2})$. Take, for example, a boost in this representation; it is given by

$$\psi \to \psi' = \begin{pmatrix} e^{\frac{\phi}{2}\sigma} & 0 \\ 0 & e^{-\frac{\phi}{2}\sigma} \end{pmatrix} \begin{pmatrix} \chi_L \\ \varphi_R \end{pmatrix}$$

If a Dirac Spinor transforms according to the $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ representation, the parity transformed object will then transform according to the $(0,\frac{1}{2})\oplus(\frac{1}{2},0)$ representation:

$$\psi^P \to (\psi^P)' = \Lambda_{(0,\frac{1}{2})\oplus(\frac{1}{2},0)}\psi^P = \begin{pmatrix} \Lambda_{(0,\frac{1}{2})} & 0 \\ 0 & \Lambda_{(\frac{1}{2},0)} \end{pmatrix} \begin{pmatrix} \varphi_R \\ \chi_L \end{pmatrix}$$

$$\therefore \psi = \begin{pmatrix} \chi_L \\ \varphi_R \end{pmatrix} \longleftrightarrow \psi^P = \begin{pmatrix} \varphi_R \\ \chi_L \end{pmatrix}$$

5.6.9 Spinors and Charge Conjugation

Earlier, we derived a transformation that yields $\chi_L \to \chi_R$ and $\varphi_R \to \varphi_L$, by the conjugation transformation. Let us view how a Dirac spinor transforms under this conjugation transformation:

$$\psi = \begin{pmatrix} \chi_L \\ \varphi_R \end{pmatrix} \to \tilde{\psi} = \begin{pmatrix} \chi_R^C \\ \varphi_R^C \end{pmatrix} = \begin{pmatrix} \chi_R \\ \varphi_L \end{pmatrix}$$

This object does not transform under boosts/Lorentz transformations as a Dirac spinor. So to transform such an object such that it maps to a Dirac spinor, it must swap our then conjugate our spinors:

$$\psi = \begin{pmatrix} \chi_L \\ \varphi_R \end{pmatrix} \to \psi^C = \begin{pmatrix} \varphi_R^C \\ \chi_L^C \end{pmatrix} = \begin{pmatrix} \varphi_L \\ \chi_R \end{pmatrix}$$

As foreshadowing, this charge conjugation operator flips all the labels that we use to describe fundamental particles.

5.6.10 On Higher Dimension Representations

We could continue to derive higher-dimensional representations of the Lorentz group, but for the purposes of this paper, it would be pointless. If curious, the table below has some further information for small irreducible representations of our group:

	m = 0	$\frac{1}{2}$	1	$\frac{3}{2}$
n = 0	Scalar	Left-chiral Weyl spinor	Self-dual 2-form	dim=4
$\frac{1}{2}$	Right-chiral Weyl spinor	4-vector	dim=6	dim=8
1	Anti-self-dual 2-form	dim=6	Traceless symmetric tensor	dim=12
$\frac{3}{2}$	dim=4	dim=8	dim=12	dim=16

Further, we will discuss some common representations of the Lorentz group below:
Irreducible Representations:

- (0,0): The one-dimensional trivial representation carried by relativistic scalar fields
- $(0,\frac{1}{2})$ or $(\frac{1}{2},0)$: The two-dimensional representations defining transformations of Fermionic super symmetry generators (spinor representation)
- $(\frac{1}{2},\frac{1}{2})$: Transformation of the four-momentum of a (massive/massless) particle (vector representation)
- $(1,1)$: Explains traceless symmetric field tensors: e.g., the traceless component of the energy-momentum tensor $T^{\mu\nu}$ (traceless rank-two tensor representation)

Off-Diagonal Direct Sums:

- $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$: The Bispinor representation of Dirac spinors

- $(1, \frac{1}{2}) \oplus (\frac{1}{2}, 1)$: The Rarita-Schwinger field representation
- $(\frac{3}{2}, 0) \oplus (0, \frac{3}{2})$: The symmetry of the hypothetical gravitino, the gauge fermion supersymmetric partner of the hypothetical graviton
- $(1, 0) \oplus (0, 1)$: The representation of a parity-invariant two-form field (the curvature form). This is the representation that the Electromagnetic Field Tensor transforms under

5.6.11 Infinite-Dimensional Representations

Thus far, we have dealt with transformations of the form: $\phi_a \to \phi'_a = M_{ab}(\Lambda)\phi_b$, where $M_{ab}(\Lambda)$ denotes the matrix of the finite-dimensional representation of the Lorentz transformation, acting on ϕ_b which is our object (e.g., spinors or vectors).

If our object ϕ were to change in space time, it would be a function of the space-time coordinates $\phi(t, x, y, z)$. Our coordinates, t, x, y, z, would also be affected by our Lorentz transformation: $\phi_a(t, x, y, z) \to M_{ab}(\Lambda)\phi_b(\Lambda(t, x, y, z))$. Two major parts give our transformation: our matrix $M_{ab}(\Lambda)$ which acts on $\phi_a(t, x, y, z)$ and the transformation of our coordinates $\Lambda(t, x, y, z)$.

The infinite-dimensional representation of the Lorentz group is given by $M^{inf}_{\mu\nu} = i(x^\mu \partial^\nu - x^\nu \partial^\mu)$, where $\partial^\nu = \frac{\partial}{\partial x_\nu}$ and x is the short hand for our coordinates. The transformation of our coordinate system can be written as $\phi'(\Lambda x) = e^{-i\frac{\omega}{2} M^{inf}_{\mu\nu}} \phi(x)$, where $\theta_i = \frac{1}{2}\epsilon_{ijk}\omega_{jk}$ and $\phi_i = i\omega_0$. The complete transformation $M_{\mu\nu} = M^{inf}_{\mu\nu} + M^{fin}_{\mu\nu}$ is

$$\phi_a(x) \to (e^{-i\frac{\omega}{2} M_{\mu\nu}})^b_a \phi_b(x)$$

With this, we have discussed all significant representations of the Lorentz group. So we will now discuss translations, the missing key to discussing the Poincaré group.

5.7 Translations

When considering an infinitesimal translation of a function along some axis, we will consider the linear expansion of the Taylor Series:

$$\phi \to \phi(x + \epsilon) = \phi(x) + \partial_x \phi(x)\epsilon$$

Since it is conventional to define the generator with an additional $-i$ factor, $P_i \equiv -i\partial_i$. Therefore, an arbitrary translation is expressed as

$$\phi(x) \to \phi(x + a) = e^{-ia^i P_i} = e^{-a^i \partial_i}$$

where a_i is our scaling factor in some direction. This is shown in the Taylor Expansion of our matrix exponential.

5.8 The Poincaré Group

Now that the cards have been set into place, let us discuss the quintessential symmetry group of physics: the Poincaré group. It is an extension of the Lorentz group, which encompasses boosts and rotations; the Poincaré group includes both and adds a third, translations. The generators of the group are J_i, K_i, P_i, which describe rotations, boosts, and translations, respectively. They interact in the Lie algebra as follows:

- $[J_i, J_j] = i\epsilon_{ijk}J_k$
- $[J_i, K_j] = i\epsilon_{ijk}K_k$
- $[K_i, K_j] = -i\epsilon_{ijk}J_k$
- $[J_i, P_j] = i\epsilon_{ijk}P_k$
- $[J_i, P_0] = 0$
- $[K_i, P_j] = i\delta_{ij}P_0$
- $[K_i, P_0] = -iP_i$

Because this is tedious, it is conventional to write in terms of $M_{\mu\nu}; J_i = \frac{1}{2}\epsilon_{ijk}M_{jk}, K_i = M_{0i}$. Hence, we have the following rules:

$$[P_\mu, P_\nu] = 0$$

$$[M_{\mu\nu}, P_\rho] = i(\eta_{\mu\rho}P_\nu - \eta_{\nu\rho}P_\mu)$$
$$[M_{\mu\nu}, M_{\rho\sigma}] = i(\eta_{\mu\rho}M_{\nu\sigma} - \eta_{\mu\sigma}M_{\nu\rho} - \eta_{\nu\rho}M_{\mu\sigma} + \eta_{\nu\sigma}M_{\mu\rho})$$

For this quite complicated group, it is vital to label our representations by using the fixed scalar values of the Casimir operators. The first being $P_\mu P^\mu := m^2$. While inconspicuous, this particle is, in actuality, the mass of a particle. The second Casimir operator is $W_\mu W^\mu$, where W^μ is the Pauli-Lubanaski pseudovector. In an extraneous computation, we can use $j \equiv j_1 + j_2$ as a label. This is known to be spin.

As we had for the Lorentz group, we have one j_i value for each representation of the $SU(2)$ algebra. For example, the (0,0) representation is spin 0, the $(\frac{1}{2}, 0), (0, \frac{1}{2})$ is spin $\frac{1}{2}$, and so on. The main takeaway is that each representation is labeled by two quantities: m and j. While m is mostly arbitrary, like for the Lorentz group, j is restricted to half-integer values.

6 A Revised Look at Elementary Particles

The labels for the irreducible representations of the Poincaré group, mass (m) and spin (j), are how particles are labeled in physics. In addition to our two labels, there exist other labels for the irreducible representations of the double cover called charges: electric charge, weak charge, and strong charge.

For right now, we can, at the very least, categorize our subatomic particles by the representations that they transform under:

- Spin 0 is described by an object, ϕ, called scalars that transform according to the $(0,0)$ representation, or Spin 0 representation (also known as the scalar representation). A notable example of this is the Higgs-boson, which is described by a scalar field.
- Spin $\frac{1}{2}$ is described by an object, ψ, called spinors that transform according to the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation, or the Spin $\frac{1}{2}$ representation (also known as the spinor representation). Most fermions transform under this representation, such as electrons and quarks.
- Spin 1 is described by an object, v, called a vector that transform under the $(\frac{1}{2}, \frac{1}{2})$ representation, or the Spin 1 representation (also known as the vector representation). Arguably, he most notable particle in this category is the photon.

What we get from deriving the irreducible representations of the Poincaré group are the mathematical tools we need to describe all elementary particles, and it serves as the foundation for all of physics.

6.1 The Label Spin

We briefly discussed one of the two labels of the Poincarè group: spin. Naturally, the other label, mass, is one of the most important quantities in physics, so what makes spin so necessary that we include it as a label along with mass? Spin is often thought of as a sort of pseudo-angular momentum, acting as an intrinsic property of a particle, much like how electric charge is. However, our particles do not, as the namesake suggests, spin or rotate with the classical sense of angular momentum. Rather, spin relies on an understanding of the internal symmetries of a particle, which in turn relies on the geometry of spatial rotations.

To begin, let us first understand the first recorded discovery of spin. In quantum physics, the Stern-Gerlach experiment demonstrated that there was a fundamental quantized property that had not yet been observed. The experiment consisted of sending electrically neutral silver atoms through an inhomogeneous magnetic field using a beam emitter, then observing their deflection onto a metallic plate. The inhomogeneous magnetic field caused the beam to split into two separate directions, creating two distinct lines on the plate. What was observed was that precisely half of the silver atoms were deflected toward the top of the metallic plate, and the other half were deflected toward the bottom. This was the first measurement of the quantum observable, and it opened many further investigations into the quantized property of spin. In 1925, George Uhlenbeck and Samuel Goudsmit proposed the first model of electron spin, stating that it obeys the principles of probabilistic superposition.

As physicists, we know nothing about the appearance or structure of particles. However, we do know that they can adopt different states depending on their orientation. Therefore, to study the spin of particles, we must define a new abstract space known as the state space. Every fundamental

particle is associated with its own quantum field, which encompasses a unique state space. Our spin number determines the behavior of the state space. The dimension of the state space is given by the expression $n = 2s + 1$, and we can express it as an nth-dimensional Hilbert space: $\mathcal{H}_n = span\{|s, m\rangle : m \in \{s, s-1, ..., -s+1, -s\}\}$, where each basis vector $|s, m\rangle$ satisfies $\hat{S}^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle$ and $\hat{S}_z |s, m\rangle = \hbar m |s, m\rangle$ ($\hat{S}$ and $\hat{S}_z$ are the spin operator and spin-z operator which are determined experimentally). However, for the sake of this paper, the full theoretical derivation is largely left out.

This aspect of spin is at the forefront of its discussion in the general physics community. However, our derivation of the spin label reveals a deeper connection between how a particle transforms and its spin beyond discussions of spin states. Hopefully, this section clarified why this label is invaluable in our discussion of physics.

6.2 Noether's Theorem

Noether's Theorem, named after Emmy Noether, was a theory published in 1918 that states that each symmetry of the Lagrangian, in physics, is related to a conserved quantity. That is, the laws of conservation of energy and momentum (as well as a quantity known as boost invariance) are directly connected to a symmetry in our derived Poincaré group. To derive a proper formulation of this, we must consider the object known as the Lagrangian.

The Lagrangian at some given instant is a scalar that describes a system's state by summarizing its potential and kinetic energy: $\mathcal{L} = T - V$, where T is the kinetic energy of a system and V is the potential. We can thus represent it as a function with the parameters of position, time, and velocity: $\mathcal{L}(x(t), \dot{x}(t), t)$. This is the case for particle theory, but we may define a similar Lagrangian function for field theories: $\mathcal{L}(\phi(\vec{x}, t), \partial_\mu \phi(\vec{x}, t), \vec{x}, t)$. For the sake of this paper, we will restrict our gaze solely to particle physics.

By the Principle of Least Action, we want to find the minimum of some functional, $S[x(t)] = \int \mathcal{L}(x(t), \dot{x}(t), t) dt$, to find the paths of our particles. This introduces the study of the calculus of variation and the famous Euler-Lagrange Equation: $\frac{\partial \mathcal{L}(x, \dot{x}, t)}{\partial x} = \frac{d}{dt}(\frac{\partial \mathcal{L}(x, \dot{x}, t)}{\partial \dot{x}})$.

With these notions set, let us begin sketching derivations of the proof of the theorem. Consider the invariance of the action under a small transformation $x \to x + \delta x$; this yields

$$\delta S = \int \delta \mathcal{L} dt = \int (\mathcal{L}(x, \frac{dx}{dt}, t) - \mathcal{L}(x + \delta x, \frac{d(x + \delta x)}{dt}, t)) dt = 0$$

For the Lagrangian to change while the action stays the same, we can add the total time derivative of some function $f(x)$ to the Lagrangian: $\mathcal{L} \to \mathcal{L} + \frac{df(x)}{dt}$, since $\delta S' - \delta S = \int_{t1}^{t2} \frac{d}{dt} \delta f(x) dt = \frac{\partial f(x)}{\partial x} \delta x|_{t1}^{t2}$ vanishes for any t1 and t2. Hence, $\delta \mathcal{L} = \frac{df(x)}{dt}$.

Let us now look at the linear expansion of the Taylor series of $\delta \mathcal{L}$:

$$\delta \mathcal{L} = -\frac{\partial \mathcal{L}}{dx} \delta x - \frac{\partial \mathcal{L}}{d\dot{x}} \delta \dot{x} = \frac{df(x)}{dt} = -\frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x) \implies \frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x + f(x)) = 0$$

Thus, the quantity $G = \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta x + f(x)$ is a conserved quantity across any interval of time.

First, let us consider a system in which there is no external force acting on our system (that is, no potential): $\mathcal{L}(t) = T - V = \frac{1}{2} m \vec{x}(t)^2$. Our Lagrangian is invariant ($\delta \mathcal{L} = 0$) under spatial translations $\vec{x(t)} \to \vec{x}(t) + \vec{a}$, where $\vec{a}$ is some positional displacement. This implies $G = m\vec{\dot{x}}(t) \cdot \vec{a} + f(x)$. Since our Lagrangian is trivially invariant under this transformation, $f(x) = 0$, so $G = m\vec{\dot{x}}(t) \cdot \vec{a}$. However, because $\frac{d}{dt} G = 0$ is true for arbitrary $\vec{a}$, our conserved quantity is $m\vec{\dot{x}}(t)$. This is the classical quantity of momentum. Hence, we have derived the law of conservation of momentum, which holds in all spacial dimensions.

We can repeat this process, but proceed with the remaining transformations in the Poincaré group. For rotations, we have $x_i(t) \to x_i(t) + \epsilon_{ijk} x_j a_k$. Again, our Lagrangian does not change under this transformation, so $f(x) = 0$, and, consequently, $G = m\dot{x}_i(t) \epsilon_{ijk} x_j a_k = (m\vec{\dot{x}}_i(t) \times \vec{x}_j(t)) \cdot \vec{a}$. Again, since $\frac{d}{dt} G = 0$ for all $\vec{a}$, our true conserved quantity is $(m\vec{\dot{x}}_i(t) \times \vec{x}_j(t))$. This is angular momentum in classical mechanics.

For time displacements, $t \to t + \Delta t$, we have that $\delta \mathcal{L} = \mathcal{L}(x(t), \dot{x}(t), t) - \mathcal{L}(x(t + \Delta t), \dot{x}(t + \Delta t), t + \Delta t) = -\frac{d\mathcal{L}}{dt} \Delta t$. Hence, $f(x) = -\mathcal{L}$ to satisfy $\delta \mathcal{L} = \frac{df(x)}{dt}$. Thus, $G = \frac{\partial \mathcal{L}}{\partial \dot{x}(t)} \dot{x}(t) - \mathcal{L} = \mathcal{H}$. Thus, our

conserved quantity is the Hamiltonian operator, representing the total energy of the system. Hence, for some change in time, energy must be conserved.

The final conserved quantity is that of the boosts (transforming to some reference frame moving with velocity v): $G = \frac{\partial \mathcal{L}}{\partial \dot{x}(t)} t - \frac{1}{2} mvt - mx(t)$. This quantity is peculiar as no well-known conservation law connects to it. Further, it is strange as it relies on some time t. However, by setting t=0 (which is permitted given G is conserved in time), we see that $G = mx(t)$. This means the center of mass of an isolated system moves at a constant velocity (linearly) through changing reference frames. This result, however, isn't necessarily an important conclusion in the scheme of our theory.

What we have proved here is an important result in particle physics and all of physics, for that matter. We have shown the underlying result between our derived Poincaré group and the conservation laws that dictate the mechanics of the universe. This principle extends to internal symmetries as well, such as gauge symmetries, which underlie the Standard Model of particle physics and lead to the conservation of quantities like charge. Noether's Theorem plays a central role in Lagrangian and Hamiltonian mechanics, quantum field theory, and general relativity, offering a role in unifying these frameworks. It shows that conservation laws are not merely axioms but rather necessary consequences of underlying symmetries in nature.

7 Conclusion

Now that we have covered the mathematical formalism, one may be inclined to ask, "What now?" Now that we have discussed Noether's Theorem, we may begin deriving the equations of nature. To do so, we must discuss the foundations of Free and Interaction Theory. These subjects allow us to determine, as the name suggests, how particles and fields interact with each other according to the representations we derived. These theories pave the way for the discussion of the entirety of physics from Quantum Mechanics and Field Theory to Classical Mechanics and Electrodynamics.

With these, we can construct the standard model, which serves as the unification of the three gauge symmetries. Unfortunately, there is one looming issue: gravity. We currently have two incompatible models of physics: Quantum Field Theory and General Relativity. Both are beyond effective at describing their respective physics, but when meshed, our intricate theories quickly unravel. This, however, should not serve as a deterrent. If anything, it should motivate us to solve the problems that the mathematicians, physicists, and scientists couldn't solve before us. While lengthy, I hope this suffices as a brief primer to particle physics, and with this, I would like to thank you for reading.

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